Differential Equations and Slope Fields

ANSWER KEY

Section 1: Slope Fields & Solution Curves

1 In the slope field for $\frac{dy}{dx} = x - y$, what is the slope at the point $(3, 1)$?

a. 2 ✓

b. 4

c. -2

d. 3

Substitute coordinates: $\frac{dy}{dx} = 3 - 1 = 2$.

2 Along which line in the slope field for $\frac{dy}{dx} = 2x + y$ are all slope segments horizontal?

a. $y = -2x$ ✓

b. $y = 2x$

c. $x = 0$

d. $y = 0$

Horizontal slopes have $\frac{dy}{dx} = 0 \Rightarrow 2x + y = 0 \Rightarrow y = -2x$.

3 If a slope field has parallel slopes along every vertical line ($x = c$), which type of differential equation could generate it?

a. $\frac{dy}{dx} = f(x)$ ✓

b. $\frac{dy}{dx} = f(y)$

c. $\frac{dy}{dx} = x + y$

d. $\frac{dy}{dx} = xy$

If slopes depend solely on x (constant for any fixed x), the differential equation is of the form $\frac{dy}{dx} = f(x)$.

Section 2: Separation of Variables

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4 Solve the differential equation $\frac{dy}{dx} = 3x^2y$ with initial condition $y(0) = 5$.

a. $y = 5e^{x^3}$ ✓

b. $y = e^{x^3} + 4$

c. $y = 5e^{3x^2}$

d. $y = x^3 + 5$

Separate variables: $\frac{1}{y}dy = 3x^2dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 5$, $A = 5$, giving $y = 5e^{x^3}$.

5 Solve $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -3$.

a. $y = -\sqrt{x^2 + 9}$ ✓

b. $y = \sqrt{x^2 + 9}$

c. $y = x^2 - 3$

d. $y = -\sqrt{x^2 - 9}$

$ydy = xdx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Since $y(0) = -3$, $9 = 2C$. Taking the negative square root to match $y(0) = -3$: $y = -\sqrt{x^2 + 9}$.

6 What is the domain of the particular solution $y = \frac{1}{2 - x}$ with $y(0) = \frac{1}{2}$?

a. $(-\infty, 2)$ ✓

b. $(2, \infty)$

c. $x \neq 2$

d. All real numbers

The solution must be on an open interval containing the initial point $x_0 = 0$. Since the discontinuity is at $x = 2$, the domain is $(-\infty, 2)$.

Section 3: Exponential Growth & Decay Models

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7 Which differential equation models a quantity whose rate of growth is directly proportional to its current size?

a. $\frac{dy}{dt} = ky$ ✓

b. $\frac{dy}{dt} = k$

c. $\frac{dy}{dt} = kt$

d. $\frac{dy}{dt} = k(M - y)$

The classic natural growth model states $\frac{dy}{dt} = ky$, whose solution is $y(t) = y_0 e^{kt}$.

8 A population grows according to $\frac{dP}{dt} = 0.05P$. If initial population is $1,000$, what is the population at $t = 20$?

a. $1000e$ ✓

b. $1000e^2$

c. 2000

d. $1000(1.05)^{20}$

$P(t) = 1000e^{0.05t}$. For $t = 20$: $P(20) = 1000e^{0.05(20)} = 1000e^1 = 1000e$.

9 A radioactive substance has a half-life of 10 years. What is its decay constant k in $\frac{dy}{dt} = -ky$?

a. $k = \frac{\ln 2}{10}$ ✓

b. $k = 10 \ln 2$

c. $k = \frac{1}{10}$

d. $k = 0.5$

$e^{-10k} = \frac{1}{2} \Rightarrow -10k = -\ln 2 \Rightarrow k = \frac{\ln 2}{10}$.

Section 4: Newton's Law of Cooling

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10 Newton's Law of Cooling states that $\frac{dT}{dt} = -k(T - T_m)$. What does T_m represent?

a. **The ambient temperature of the surrounding medium ✓**

b. The initial temperature of the object

c. The maximum temperature reached

d. The rate of heat transfer

In Newton's Law of Cooling, T_m is the constant ambient temperature of the surrounding environment.

11 A cup of coffee at $90\,^\circ$C cools in a room of $20\,^\circ$C. If its temperature after 10 minutes is $60\,^\circ$C, what is the temperature difference $T - 20$ at $t = 10$?

a. $40\,^\circ$**C ✓**

b. $70\,^\circ$C

c. $30\,^\circ$C

d. $20\,^\circ$C

Temperature difference is $60 - 20 = 40\,^\circ$C.

12 What is $\lim_{t \to \infty} T(t)$ in Newton's Law of Cooling?

a. T_m ✓

b. 0

c. ∞

d. T_0

As time approaches infinity, the object's temperature asymptotically approaches the surrounding temperature T_m.

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